How-To Library

Informatica

Performance Tuning and Sizing Guidelines for PowerExchange for Db2 Warehouse for PowerCenter

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Abstract

Use PowerExchange for Db2 Warehouse to learn how to read from and write to Db2 Warehouse. Multiple factors such as data set size, hardware parameters, and mapping parameters impact the adapter performance. To optimize the performance, you can analyze your data set size, use the recommended hardware, and tune these parameters appropriately.

Supported Versions

- PowerExchange for Db2 Warehouse for PowerCenter 10.4.1

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Overview

Performance tuning is an iterative process where you analyze the performance, use guidelines to estimate and define parameters that impact the performance, monitor, and adjust the results as required. Sizing is a process where you analyze the data set and use the recommended hardware to improve the performance.

This document describes the key data size, hardware, and Informatica mapping parameters that you can tune to optimize the performance of PowerExchange for Db2 Warehouse for PowerCenter.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of PowerExchange for Db2 Warehouse might vary based on individual environments and other parameters even when you use the same data.

Performance Sizing Recommendations

You can tune the hardware parameters based on the data set and use the recommended hardware capacities to optimize the performance of the machine where the PowerCenter Integration Service runs. Performance sizing recommendations depend on the size of the data set.

Consider the sizing recommendations for the following types of data set:

Sandbox data set

A data set is considered to be a sandbox data set if you read or write up to 7.5 GB of data.

For a sandbox data set, you can read up to 7.5 GB of data from a local file system to Db2 Warehouse in `0:11:00(HH:MM:SS)` minutes and from Db2 Warehouse to a flat file in `0:06:40(HH:MM:SS)` minutes.

The following table describes the sizing recommendations for the system that processes the sandbox data sets:

Operation	Writer	Reader
Services	PowerCenter Integration Service	PowerCenter Integration Service
VCPUs	2 (1+1) (PMDTM + Domain)	2 (1+1) (PMDTM + Domain)
Memory	1.15 GB (0.15+1) (PMDTM + Domain)	1.15 GB (0.15+1) (PMDTM + Domain)
Storage	40 GB	40 GB

Medium data set

A data set is considered to be a medium data set if you read or write up to 250 GB of data.

For a medium data set, you can read up to 250 GB of data from a local file system to Db2 Warehouse in `01:00:00(HH:MM:SS)` hours and from Db2 Warehouse to a flat file in `01:00:00(HH:MM:SS)` hours by using key range partitioning.

The number of partitions recommended for reading and writing data to Db2 Warehouse is 6.

The following table describes the sizing recommendations for the system that processes medium data sets:

Operation	Writer	Reader
Services	PowerCenter Integration Service	PowerCenter Integration Service
VCPUs	5 (4+1) (PMDTM + Domain)	7 (6+1) (PMDTM + Domain)
Memory	1.75 GB (0.75+1) (PMDTM + Domain)	1.75 GB (0.75+1) (PMDTM + Domain)
Storage	270 GB	270 GB

Large data set

A data set is considered to be a large data set if you read or write up to 500 GB of data.

For a large data set, you can read up to 500 GB of data from a local file system to Db2 Warehouse in `01:00:00(HH:MM:SS)` hours and from Db2 Warehouse to a flat file in `01:00:00(HH:MM:SS)` hours.

The number of partitions recommended for writing data to Db2 Warehouse is 13 and for reading data from Db2 Warehouse is 12.

The following table describes the sizing recommendations for the system that processes large data sets:

Operation	Writer	Reader
Services	PowerCenter Integration Service	PowerCenter Integration Service
VCPUs	8 (7+1) (PMDTM + Domain)	13 (12+1) (PMDTM + Domain)

Operation	Writer	Reader
Memory	2.4 GB (1.4+1) (PMDTM + Domain)	4 GB (3+1) (PMDTM + Domain)
Storage	520 GB	520 GB

Performance Tuning Parameters

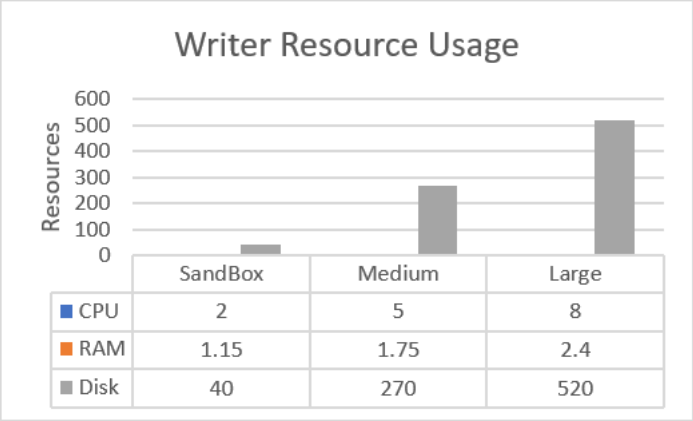
You can optimize the performance of Db2 Warehouse mappings by tuning the following parameters:

- Hardware
- PowerCenter Integration Service
- PowerCenter Mapping

Tune the Hardware

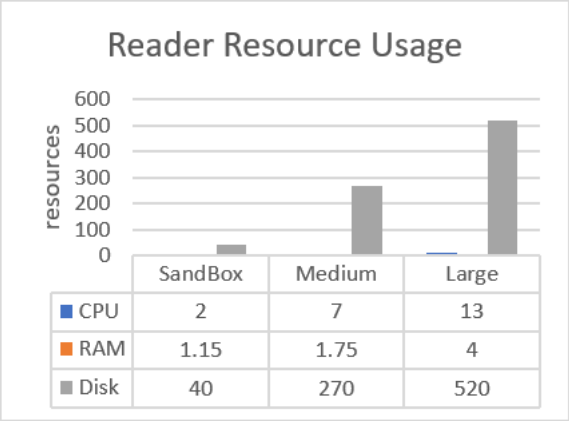
You can tune the hardware parameters to optimize the performance of the machine where the PowerCenter Integration Service runs.

The following graph illustrates the writer resource usage based on the hardware requirements of the data set:



The hardware requirements change with respect to change in data volume and individual component settings in the writer.

The following graph illustrates the reader resource usage based on the hardware requirements of the data set:



The hardware requirements change with respect to change in data volume and individual component settings in the reader.

Instance Size for PowerCenter Integration Service

When you provision an Amazon local instance, you must choose the instance size of your nodes appropriately as some workloads are CPU intensive while others are disk I/O or memory intensive.

Use the following guidelines to choose the right instance size for PowerCenter Integration Service in an Amazon local instance:

- For memory-intensive jobs, you must ensure that the `m2` family instance size has enough memory and CPU power to perform the task.
- For CPU-intensive jobs, you must ensure that you have a `c2-series` instance.
- If the jobs are both memory and CPU intensive, you must ensure that you use cg1.4xlarge instance or cc2.8xlarge instance that have suitable memory and CPU power to handle a heavy workload.

Tune the PowerCenter Integration Service

You can tune the PowerCenter Integration Service to optimize the performance of the Db2 Warehouse components by choosing the Db2 Warehouse in the same region as the Informatica PowerCenter machine and by selecting the right instance size for Informatica PowerCenter.

Tune the Mapping

You can tune the parameters in the mapping to optimize the performance.

The following table lists the tuning recommendations for the read and write mapping parameters according to the size of the data set:

Mapping Parameter	Sandbox Data Set	Medium Data Set	Large Data Set
Buffer Block Size	Auto	Auto	Auto
DTM Buffer Size	Auto	Auto	Auto

Buffer Block Size

The buffer block size determines the size of the blocks of data used to move rows of data from the source to the target. You can specify the buffer block size in bytes or a percentage of total memory.

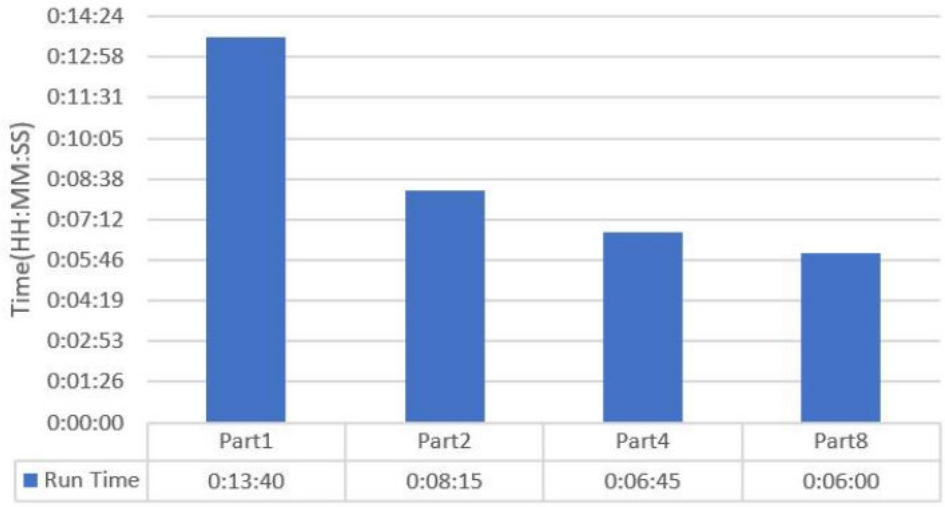
DTM Buffer Size

The DTM buffer size determines the size of the buffer blocks that the PowerCenter Integration Service uses to move data from sources to targets. All the buffer blocks are allocated from the DTM buffer pool.

Partitions

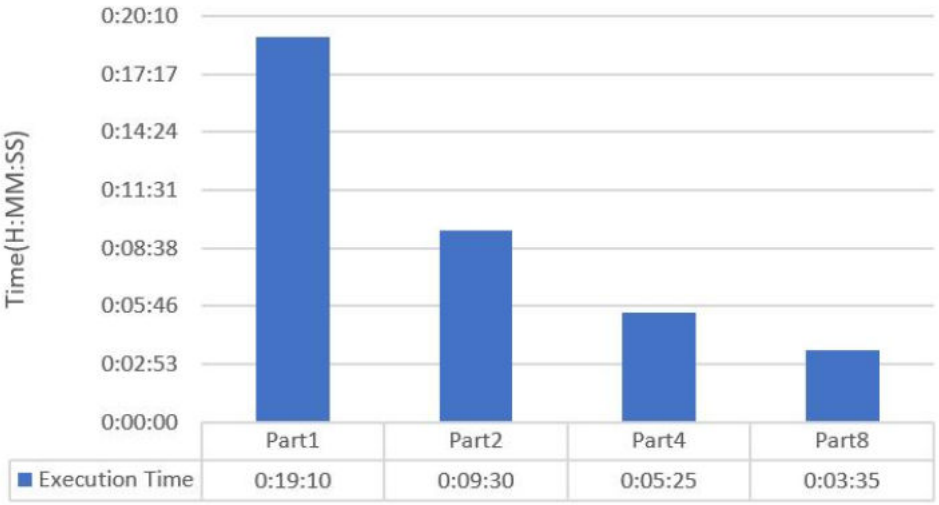
You can configure partitions to optimize the mapping performance. Specify the number of partitions configured for the read or write operation in each mapping.

The following graph illustrates the impact of partitioning on the reader performance:



When you configure partitioning for the read operation, the reader performance improves four times as compared to without partitioning.

The following graph illustrates the impact of partitioning on the writer performance:



When you configure partitioning for the write operation, the writer performance improves two to four times as compared to without partitioning.

Author

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